

IDC 410: Machine Learning

Spring Semester 2025-26

Assignment 08

Instructor: Prof. Shashikant R. Dugad

Uploaded on 18th March 2026 (Wednesday)

Submission: Before 26th March 2026 (Thursday) 23:00 hrs

1. Locating ball (Inside or Outside the Fence) with new dataset as discussed in class
You are provided with two folders (../Fence/Inside and ../Fence/Outside)
 - A) Observe each image in both folders carefully and make sure that they represent the correct class. You may reject those images that are not very clear or have been wrongly labeled.
 - B) Develop the CNN for identifying the ball if it is inside or outside the fence. Describe details of architecture of the network that you used for training and validation.
 - C) Provide detailed performance of BOTH the network on validation data.
2. Construct CNN architecture to identify numbers in MNIST dataset. Describe the architecture and results obtained using this network. Compare the performance with FCN architecture used earlier

Please NOTE:

a) Complete your assignment in time

b) If you need any clarifications, feel free to contact TA's or me well in advance so that you can complete your assignment in time